

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT1 — RELATIONSHIP VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Relationship Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO4 Statement:	<i>Analyze relationships between variables using scatterplots, correlograms, PCA, bubble charts, and network graphs to construct and interpret appropriate visualizations. (BL4)</i>		
Student Name:	Jothika M	Reg. No.:	192324235
Section / Batch:	SLOT C	Date:	22/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Graph paper or a ruler is recommended for accurate, to-scale plotting.
- Show all supporting calculations (e.g., variance explained, correlation strength, node degree) where relevant.
- Every chart must include a title, correctly labeled axes, and a legend where relevant.

Annexure A – RELATIONSHIP VISUALIZATION LAB

Problem 2 (5 Marks)

Data: Study hours and exam scores for 10 students: (2,45), (3,50), (4,58), (5,62), (6,68), (7,75), (8,80), (9,85), (10,90), (1,40).

Task: Construct a scatterplot, sort the students by study hours, and identify the type of relationship shown.

Problem 9 (5 Marks)

Data: Correlation values: Rainfall–CropYield = 0.7, Rainfall–Temperature = –0.3, CropYield–Temperature = –0.5.

Task: Construct a correlogram and interpret what the negative correlation values indicate about this data.

Problem 11 (5 Marks)

Data: Height and Weight for 6 people: (150,48), (155,52), (160,58), (165,62), (170,68), (175,72).

Task: Calculate the mean of each variable and describe what PC1 would represent for this dataset.

Problem 18 (5 Marks)

Data: 6 products' Price (X, \$), Customer Rating (Y, /5), Units Sold (bubble size): P1(20,4.5,500), P2(35,3.8,300), P3(15,4.2,700), P4(50,4.8,150), P5(25,4.0,600), P6(40,3.5,250).

Task: Construct a bubble chart and identify which product has the best combination of high rating and high sales.

Problem 25 (5 Marks)

Data: A communication network of 5 departments: Dept1–Dept2, Dept2–Dept3, Dept3–Dept4, Dept4–Dept5, Dept5–Dept1.

Task: Draw the network graph and determine whether this forms a cycle (closed loop).

Rubrics for Calculation
ASSESSMENT RUBRIC
AT1 — Relationship Visualization Lab

CO4 · BT4: Analyze ·

This rubric is used to assess student responses to the CO4-AT1 Relationship Visualization Lab problems, in which students construct and analyze scatterplots, correlograms, PCA outputs, bubble charts, or network graphs from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant relationship-visualization technique (scatterplot, correlogram, PCA, bubble chart, or network graph) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned analysis of the constructed visualization, explicitly tied to the specific values in the dataset.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies construction and analysis techniques to produce a complete, accurate	Applies techniques to produce a correct visualization, with only minor errors.	Limited application; the visualization partially represents the data but has	Fails to apply appropriate techniques; the visualization does not correctly

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		visualization correctly matched to the data.		notable errors.	represent the data.
Organization	15%	The response is clearly structured, with construction and analysis presented in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.
Accuracy	10%	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 2 – Scatter Plot – Study Hours vs Exam Scores

Introduction

A **scatter plot** is used to study the relationship between two numerical variables. Each point represents one student's study hours and corresponding exam score.

Here:

- **X-axis** → Study Hours
- **Y-axis** → Exam Score

The students should first be sorted according to their study hours so that the increasing pattern can be observed clearly.

Given Data

Student	Study Hours	Exam Score
1	2	45
2	3	50
3	4	58
4	5	62
5	6	68
6	7	75
7	8	80
8	9	85
9	10	90
10	1	40

Sorted Data : After sorting according to study hours:

Study Hours	Exam Score
1	40
2	45
3	50
4	58
5	62
6	68
7	75
8	80
9	85
10	90

R Code

R

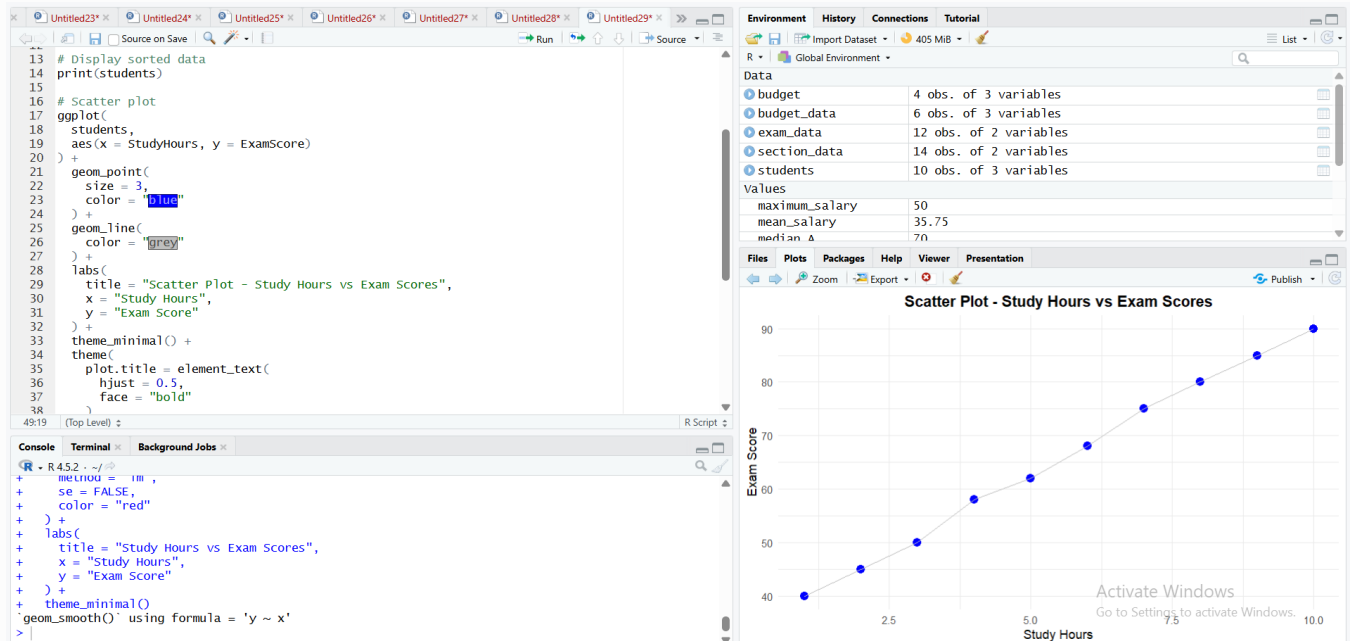
```
library(ggplot2)

# Create data
students <- data.frame(
  Student = 1:10,
  StudyHours = c(2, 3, 4, 5, 6, 7, 8, 9, 10, 1),
  ExamScore = c(45, 50, 58, 62, 68, 75, 80, 85, 90, 40)
)

# Sort students by study hours
students <- students[order(students$StudyHours), ]

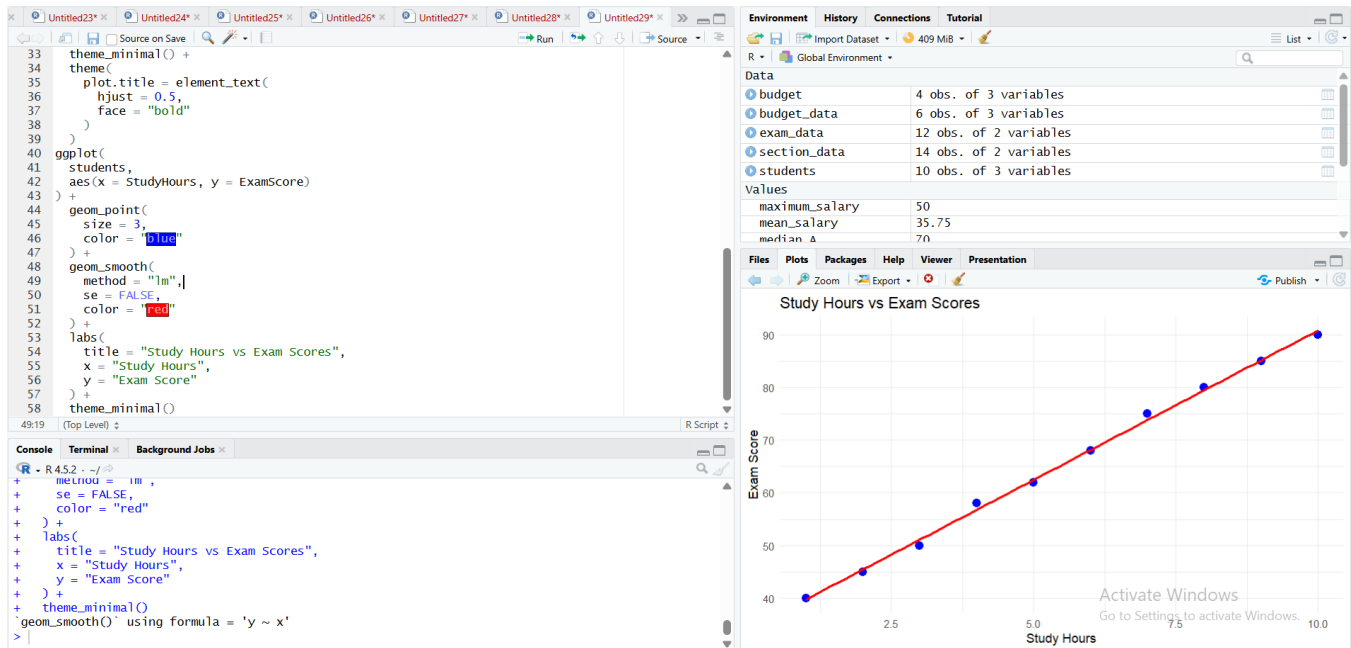
# Display sorted data
print(students)

# Scatter plot
ggplot(
  students,
  aes(x = StudyHours, y = ExamScore)
) +
  geom_point(
    size = 3,
    color = "blue"
  ) +
  geom_line(
    color = "grey"
  ) +
  labs(
    title = "Scatter Plot - Study Hours vs Exam Scores",
    x = "Study Hours",
    y = "Exam Score"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
)
```



Optional: Add Trend Line

```
R
ggplot(
  students,
  aes(x = StudyHours, y = ExamScore)
) +
  geom_point(
    size = 3,
    color = "blue"
  ) +
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "red"
  ) +
  labs(
    title = "Study Hours vs Exam Scores",
    x = "Study Hours",
    y = "Exam Score"
  ) +
  theme_minimal()
```



Evidence

The sorted data shows that as study hours increase from **1 hour to 10 hours**, exam scores increase from **40 to 90**.

Examples:

- 1 hour → 40 marks
- 5 hours → 62 marks
- 8 hours → 80 marks
- 10 hours → 90 marks

Interpretation

The scatter plot shows a **strong positive relationship** between study hours and exam scores. As the number of study hours increases, students generally achieve higher exam scores. The points follow an upward pattern from left to right.

This indicates that students who spend more time studying tend to obtain better exam results.

However, the scatter plot shows an **association**, not necessarily a direct cause-and-effect relationship. Other factors such as prior knowledge, teaching quality, and learning ability can also influence exam scores.

Conclusion

The scatter plot indicates a **strong positive relationship** between study hours and exam scores. The sorted data clearly demonstrates that higher study time is associated with higher examination performance.

Problem 9 – Correlogram – Correlation Analysis

Introduction

A **correlogram** is a graphical representation of correlation coefficients between multiple variables. It uses colors and numerical values to show the **strength and direction of relationships**.

Correlation ranges from:

$$-1 \leq r \leq +1$$

- **+1** → Perfect positive relationship
- **0** → No linear relationship
- **-1** → Perfect negative relationship

Given Data

Variable Pair	Correlation
Rainfall – Crop Yield	+0.7
Rainfall – Temperature	-0.3
Crop Yield – Temperature	-0.5

Correlation Matrix

The complete correlation matrix is:

	Rainfall	Crop Yield	Temperature
Rainfall	1.0	0.7	-0.3
Crop Yield	0.7	1.0	-0.5
Temperature	-0.3	-0.5	1.0

R Code

R

```
library(ggplot2)

# Create correlation matrix
cor_matrix <- matrix(
  c(
    1.0, 0.7, -0.3,
    0.7, 1.0, -0.5,
    -0.3, -0.5, 1.0
  ),
  nrow = 3,
  byrow = TRUE
)

colnames(cor_matrix) <- c(
  "Rainfall",
  "Crop Yield",
  "Temperature"
)

rownames(cor_matrix) <- c(
  "Rainfall",
  "Crop Yield",
  "Temperature"
)

# Convert matrix to data frame for ggplot
cor_data <- as.data.frame(as.table(cor_matrix))

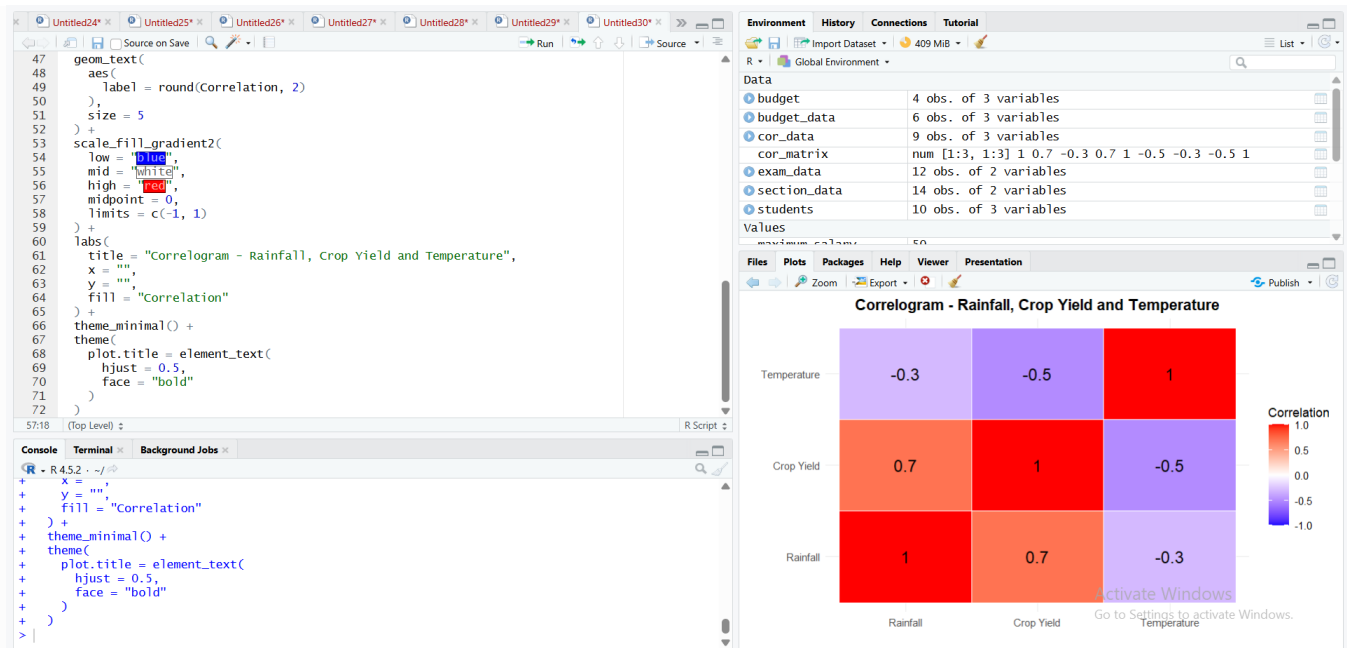
colnames(cor_data) <- c(
  "Variable1",
  "Variable2",
  "Correlation"
)

# Correlogram
ggplot(
  cor_data,
  aes(
    x = Variable1,
    y = Variable2,
```

```

    fill = Correlation
  )
) +
  geom_tile(
    color = "white"
  ) +
  geom_text(
    aes(
      label = round(Correlation, 2)
    ),
    size = 5
  ) +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    limits = c(-1, 1)
  ) +
  labs(
    title = "Correlogram - Rainfall, Crop Yield and Temperature",
    x = "",
    y = "",
    fill = "Correlation"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
)

```



Evidence

Rainfall and Crop Yield

$$r=0.7$$

This indicates a **strong positive correlation**.

Rainfall and Temperature

$$r=-0.3$$

This indicates a **weak negative correlation**.

Crop Yield and Temperature

$$r=-0.5$$

This indicates a **moderate negative correlation**.

Interpretation of Negative Correlation

A **negative correlation** means that when one variable increases, the other variable tends to decrease.

For Rainfall and Temperature:

$$r=-0.3$$

This suggests that higher rainfall is generally associated with slightly lower temperatures.

For Crop Yield and Temperature:

$$r = -0.5$$

This suggests that higher temperatures tend to be associated with lower crop yields in this dataset.

The negative value indicates the **direction** of the relationship, while the magnitude indicates its strength.

Conclusion

The correlogram shows that rainfall has a positive relationship with crop yield (**0.7**), while temperature has negative relationships with both rainfall (**-0.3**) and crop yield (**-0.5**). Negative correlations indicate that the variables tend to move in opposite directions.

Problem 11 – PCA – Height and Weight

Introduction

Principal Component Analysis (PCA) is a dimensionality-reduction technique that transforms correlated variables into a smaller number of new variables called **Principal Components**.

The first principal component (**PC1**) captures the **maximum possible variation** in the original data.

Given Data

Person	Height (cm)	Weight (kg)
1	150	48
2	155	52
3	160	58
4	165	62
5	170	68
6	175	72

Mean Calculation

Mean Height

$$6150 + 155 + 160 + 165 + 170 + 175 = 6975 = \mathbf{162.5 \text{ cm}}$$

Mean Weight

$$648+52+58+62+68+72=6360=60 \text{ kg}$$

Therefore:

- **Mean Height = 162.5 cm**
- **Mean Weight = 60 kg**

R Code

R

Create data

```
height_weight <- data.frame(  
  Height = c(150, 155, 160, 165, 170, 175),  
  Weight = c(48, 52, 58, 62, 68, 72)  
)
```

Calculate means

```
mean_height <- mean(height_weight$Height)  
mean_weight <- mean(height_weight$Weight)
```

```
cat(  
  "Mean Height:",  
  mean_height,  
  "cm\n"  
)
```

```
cat(  
  "Mean Weight:",  
  mean_weight,  
  "kg\n"  
)
```

Perform PCA

```
pca_result <- prcomp(  
  height_weight,  
  scale. = TRUE  
)
```

Display PCA result

```
print(pca_result)
```

Variance explained

```
summary(pca_result)
```

```
# PCA scores
```

```
print(pca_result$x)
```

```
+ )
>
> # Display PCA result
> print(pca_result)
Standard deviations (1, ..., p=2):
[1] 1.41364120 0.04023134

Rotation (n x k) = (2 x 2):
      PC1      PC2
Height 0.7071068 -0.7071068
Weight 0.7071068  0.7071068
>
> # Variance explained
> summary(pca_result)
Importance of components:
      PC1      PC2
Standard deviation    1.4136 0.04023
Proportion of Variance 0.9992 0.00081
Cumulative Proportion 0.9992 1.00000
>
> # PCA scores
> print(pca_result$x)
      PC1      PC2
[1,] -1.866354  0.02346851
[2,] -1.181242 -0.04734841
[3,] -0.342556  0.03540846
[4,]  0.342556 -0.03540846
[5,]  1.181242  0.04734841
[6,]  1.866354 -0.02346851
> |
```

Evidence

Height and weight both increase together:

- 150 cm → 48 kg
- 160 cm → 58 kg
- 170 cm → 68 kg
- 175 cm → 72 kg

Therefore, the two variables show a strong positive relationship.

What Does PC1 Represent?

Since **height and weight increase together**, PC1 will primarily represent the common variation associated with **overall body size**.

A person with a higher PC1 score would generally have:

- Greater height
- Greater weight

A person with a lower PC1 score would generally have:

- Lower height
- Lower weight

Therefore, PC1 can be interpreted as a general **body-size dimension** combining information from both height and weight.

Interpretation

The two original variables are highly related. Instead of analyzing height and weight separately, PCA can summarize their common variation using PC1.

PC1 captures the largest amount of variation in the dataset and represents the combined pattern of increasing height and weight.

Conclusion

The mean height is **162.5 cm**, and the mean weight is **60 kg**. Since height and weight increase together, **PC1 mainly represents overall body size**, capturing the common variation between the two variables.

Problem 18 – Bubble Chart – Price, Rating and Sales

Introduction

A **bubble chart** is an extension of a scatter plot that represents **three numerical variables simultaneously**.

- X-axis → Price
- Y-axis → Customer Rating
- Bubble Size → Units Sold

This allows price, customer satisfaction, and sales volume to be compared in one visualization.

Given Data

Product	Price (\$)	Rating (/5)	Units Sold
P1	20	4.5	500
P2	35	3.8	300
P3	15	4.2	700
P4	50	4.8	150
P5	25	4.0	600
P6	40	3.5	250

R Code

```
R
library(ggplot2)

products <- data.frame(
  Product = c(
    "P1", "P2", "P3",
    "P4", "P5", "P6"
  ),
  Price = c(
    20, 35, 15,
    50, 25, 40
  ),
  Rating = c(
    4.5, 3.8, 4.2,
    4.8, 4.0, 3.5
  ),
  Units Sold = c(
    500, 300, 700,
    150, 600, 250
  )
)
```



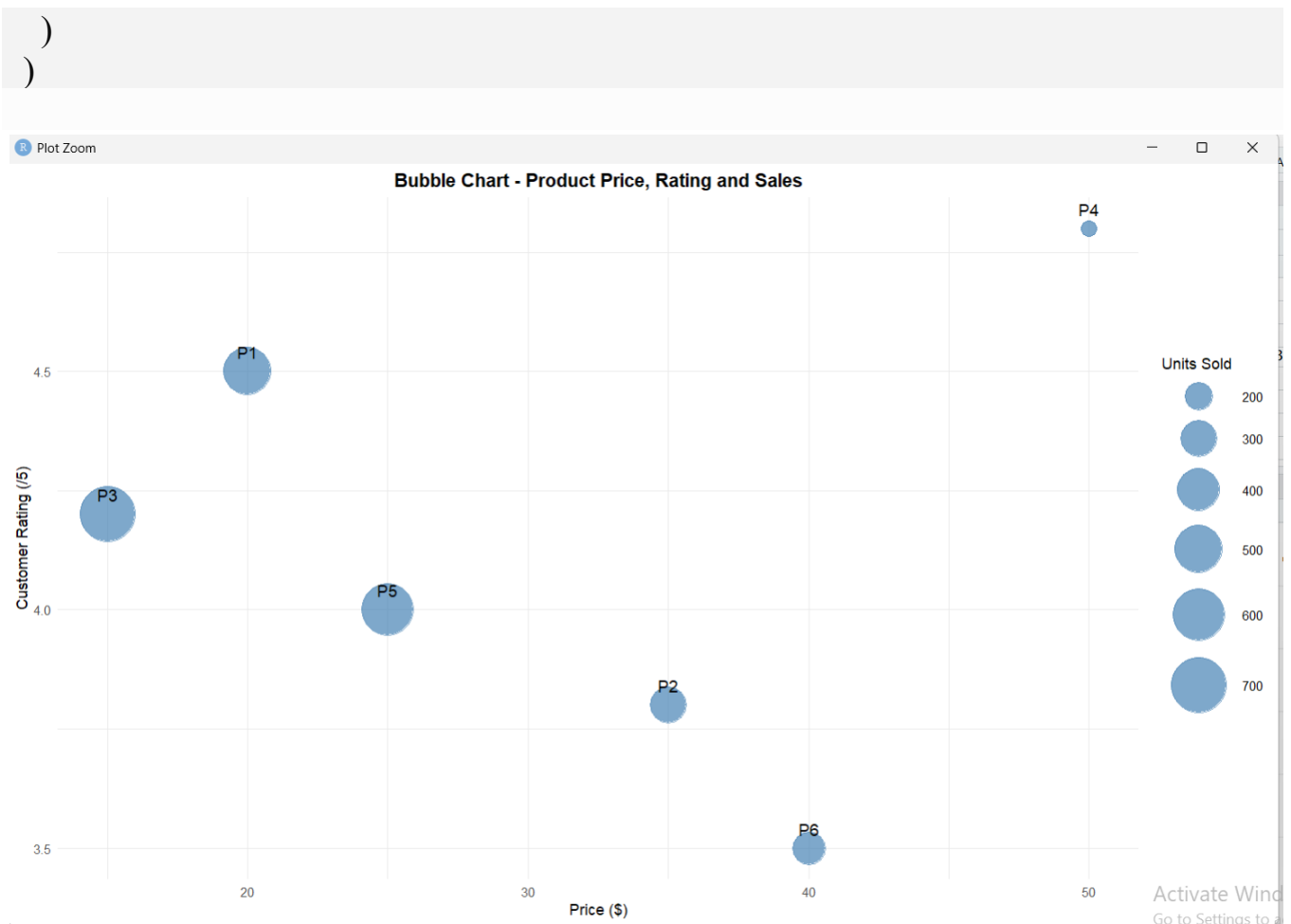
```

UnitsSold = c(
  500, 300, 700,
  150, 600, 250
)
)

# Display data
print(products)

# Bubble chart
ggplot(
  products,
  aes(
    x = Price,
    y = Rating,
    size = UnitsSold
  )
) +
  geom_point(
    alpha = 0.7,
    color = "steelblue"
  ) +
  geom_text(
    aes(label = Product),
    vjust = -1,
    size = 4
  ) +
  scale_size_continuous(
    range = c(5, 18)
  ) +
  labs(
    title = "Bubble Chart - Product Price, Rating and Sales",
    x = "Price ($)",
    y = "Customer Rating (/5)",
    size = "Units Sold"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )

```



Evidence

Highest Rating

Product P4 has the highest rating:

4.8/5

However, P4 sells only:

150 units

Highest Sales

Product P3 has the highest sales:

700 units

with a rating of:

4.2/5

Product P5

P5 has:

- Price = \$25
- Rating = 4.0
- Units Sold = 600

Interpretation

Product P3 provides the best overall combination of high rating and high sales volume.

It has the largest sales volume of 700 units and a relatively high customer rating of 4.2/5. Its price is also the lowest at \$15, which may contribute to its strong sales.

P4 has the highest rating, but its sales volume is very low. Therefore, if both customer rating and sales volume are considered together, P3 is the strongest product.

Conclusion

The bubble chart allows three variables to be compared simultaneously. **P3 is the strongest overall performer**, combining a good customer rating with the highest number of units sold.

Problem 25 – Network Graph – Communication Network

Introduction

A **network graph** is used to represent relationships or connections between entities. In a network:

- **Nodes/vertices** represent entities.
- **Edges/links** represent connections between entities.

Here, the five departments form a communication network.

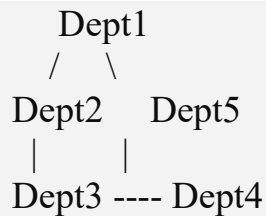
Given Connections

The network contains:

1. Dept1 – Dept2

2. Dept2 – Dept3
3. Dept3 – Dept4
4. Dept4 – Dept5
5. Dept5 – Dept1

The network can be represented as:



This forms a **closed loop**.

R Code

```
R
# Install package if required
# install.packages("igraph")

library(igraph)

# Create edges
edges <- c(
  "Dept1", "Dept2",
  "Dept2", "Dept3",
  "Dept3", "Dept4",
  "Dept4", "Dept5",
  "Dept5", "Dept1"
)

# Create network graph
network <- graph(
  edges = edges,
  directed = FALSE
)

# Display graph information
print(network)
```

```

# Draw network
plot(
  network,
  vertex.color = "lightblue",
  vertex.size = 35,
  vertex.label.color = "black",
  vertex.label.cex = 0.9,
  edge.color = "black",
  main = "Communication Network of Five Departments"
)

# Check whether the graph contains a cycle
cycles <- length(cycle_basis(network))

cat(
  "Number of cycles:",
  cycles,
  "\n"
)

```

Simpler cycle check

```

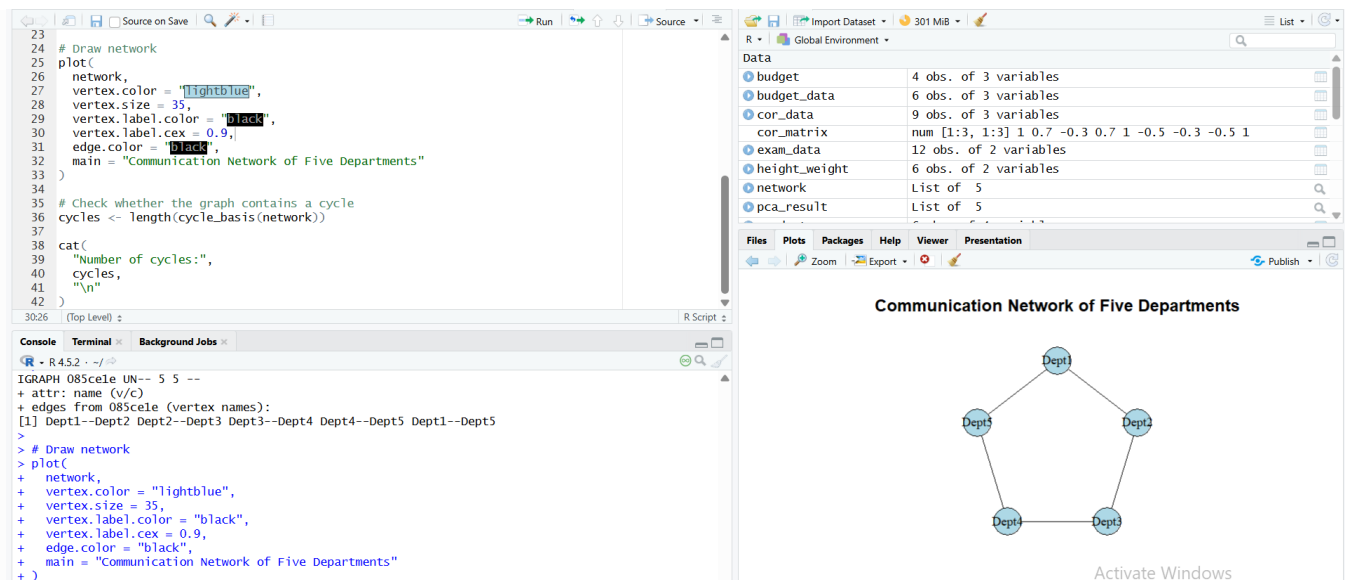
R
library(igraph)

network <- make_ring(5)

V(network)$name <- c(
  "Dept1",
  "Dept2",
  "Dept3",
  "Dept4",
  "Dept5"
)

plot(
  network,
  vertex.color = "lightblue",
  vertex.size = 35,
  vertex.label.color = "black",
  main = "Department Communication Network"
)

```



Evidence

Starting from Dept1, we can travel through:

Dept1 → Dept2 → Dept3 → Dept4 → Dept5 → Dept1

The path returns to the starting department.

Therefore, every department is connected and the network forms a closed path.

Interpretation

The network is a **cycle** because the sequence of connections starts at Dept1 and eventually returns to Dept1 without needing to repeat an intermediate department.

The network can therefore support communication around the complete loop:

Dept1 → Dept2 → Dept3 → Dept4 → Dept5 → Dept1

Conclusion

Yes, the communication network **forms a cycle (closed loop)**. All five departments are connected in a circular structure, and there is a path from each department around the network that eventually returns to the starting department.